SABASTIAN ASUO-DARKO

BIOMEDICAL ENGINEER • RESEARCHER

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personal website

Certifications: Medical Device Essentials — SternMed · WHO Covid-19 Respiratory Equipment · Sysmex Caresphere · ALX AI Career Essentials · IT Security & AI

SUMMARY

Biomedical Engineer with First-Class Honours and a peer-reviewed publication. Research experience in computational biomechanics, focusing on pulmonary mechanics and VILI using ANSYS. Strong foundation in biomaterials science, including practical lab work with bio-based polymers (alginate, chitosan) and bacterial cellulose. Seeking an MSc/PhD position to advance adaptive biomedical device innovation through simulation-driven design.

Curious, coachable, and research-driven.

EDUCATION

Kwame Nkrumah University of Science & Technology (KNUST) — Kumasi, Ghana
BSc. Biomedical Engineering – First Class Honours — Jan 2021 – Nov 2024
CWA 77.75/100 Top 10% of cohort

RESEARCH INTERESTS

- Computational Biomechanics & Physiological Modelling
- Adaptive Medical Device Design, Simulation & Validation
- Respiratory Mechanics & Ventilator-Induced Lung Injury (VILI)
- Biomaterials for Biomedical Applications
- Biomedical Systems Analysis & Translational Engineering

PUBLICATIONS & PROJECTS

Causal Factors and Their Effects on the Body for the Design of a Smart Bedsore Prevention System

Sept 2023 – Jan 2024

Co-authored Publication — IJITIS *KNUST, Ghana*

- Analyzed physiological data (pressure, temperature, humidity) to inform control logic for a smart bedsore prevention device, projecting up to a 40% reduction in incidence for high-risk patients.
- Informed the design of the prototype assistive device integrating sensor data to autonomously monitor and mitigate pressure ulcer risks.
- View Publication

Simulating Pulmonary Mechanics

Jan 2024 – Aug 2024

Undergraduate Thesis — Computational Modelling of VILI *KNUST, Ghana*

- Led a two-person project modelling spatial pressure propagation in lung tissue to study Ventilator-Induced Lung Injury (VILI).
- Built anatomically inspired 3D lung models and ran ANSYS simulations under varying ventilation pressures to optimize safe operational thresholds.
- View Documentation

Biomaterials & Biochemistry Practical Coursework

Jan 2024 – May 2024

Laboratory Training & Material Analysis *KNUST, Ghana*

- Completed biomaterials and biochemistry lab coursework involving alginate, chitosan, and bacterial cellulose.
- Gained practical exposure to hydrogel preparation, material behaviour, and material–biological interaction.
- View Lab Documentation

Design and Finite Element Analysis of an Above-Knee Prosthetic Leg

Jan 2022

Assistive Technology Design Project *KNUST, Ghana*

- Engineered a low-cost, anatomically optimized above-knee prosthetic leg, laying the groundwork for future integration of adaptive, sensor-driven prosthetic systems.
- Performed FEA in SolidWorks to assess stress, displacement, and strain under dynamic physiological loading, ensuring safety and durability.
- View Documentation

Computational Design and Testing of Sports Equipment
June 2024

Engineering Simulation Project — *KNUST, Ghana*

- Applied computational modelling to analyse stress distribution and performance characteristics.
- Used simulation-driven insights to guide iterative design improvements.
- View Documentation

Intermittent Pneumatic Compression Device for DVT Prevention
Feb 2024 – Sep 2024

Assistant Researcher · **Prototyping** — *KNUST, Ghana*

- Designed and prototyped a low-cost intermittent pneumatic compression device tailored for low-resource Ghanaian healthcare settings.
- Managed the full development cycle: clinical needs assessment, hardware design, Arduino-controlled pneumatics, SolidWorks enclosure CAD, and cost analysis.
- View Documentation

INDUSTRY & CLINICAL EXPERIENCE

CROWN HEALTHCARE — BIOMEDICAL ENGINEER

Accra, Ghana — Sep 2026 – Present

Project: IVD & General Medical Equipment • **Client:** Various Healthcare Facilities

- Install, calibrate, maintain and train on IVD analysers; ensure 90% uptime.
- Maintain Anaesthesia machines and other general medical equipment; reduce failure rates by 20%.

Equipment handled: CLIA & Chemistr Analyser (Snibe Maglumi X6, 800, Bioassay 240 Plus), , Anaesthesia (Comen AX) etc.

SKYLER HEALTHCARE / MACHINE MEDICS — FREELANCE FIELD SERVICE & SALES ENGINEER

Accra & Oyarifa, Ghana — Jan 2026 – Aug 2026

Project: IVD & General Medical Equipment • **Client:** Various Healthcare Facilities

- Installed and commissioned 30+ diagnostic devices across 20+ sites with full regulatory compliance.
- Trained 40+ clinical staff; led technical tenders securing 5 new contracts.

Equipment handled: FIA (Vacure Lyofia, iSIA Biotech), ECG (Healicom), Chemistry (Heto AU120, Genrui GS 100, Cornley Mini ISE), Haematology (Zybio Z5, Z3, XZ600)

SYSMEX WEST AND CENTRAL AFRICA — FIELD SERVICE ENGINEER (ASST.)

Cantonments, Ghana — Oct 2024 – Jan 2026

Project: IVD • **Client:** Various Healthcare Facilities

- Conducted site readiness for 20+ analysers; reduced setup delays by 30%.
- Executed monthly preventive maintenance on 20+ analysers; reduced downtime by 20%.

Equipment handled: Haematology Analyser (Sysmex XN Series, XN-L, XP, XQ), Urine Analyser (Sysmex UC, UF)

Internship — UGMC & Ministry of Health, Biomedical Engineering Intern (Sept 2023–Dec 2023 & Sept 2021–Dec 2021). Maintained 20+ devices monthly (25% failure reduction), repaired 10+ critical devices, validated 30+ across 5+ hospitals.

Clinical Engineering Portfolio: See the devices I've worked on and the impact I've delivered — **view the full portfolio**

VOLUNTEER EXPERIENCE & COMMUNITY IMPACT

- **GRAMIC's Foundation** — Video Editor (Volunteer), Remote (May 2026 – Present)

- **Methodist Church of the Resurrection, Kpeehe** — Graphic Designer (2022 – Present)
- **BMESS** — Deputy Electoral Committee Head (Oct 2023 – Oct 2024)
- **BMESS** — Deputy Judicial Committee Head (Oct 2022 – Oct 2023)
- **BMESS Day of Code** — Winner (July 2023)

SOFTWARE COMPETENCIES

Engineering & Simulation
ANSYS, MATLAB, LabVIEW

Programming & Data
Python, C, C++, SQL, LaTeX, AI & ML

Design & Analysis
SolidWorks, AutoCAD, Blender

Productivity & Reporting
Microsoft Office, Google Workspace, ZohoOne

Other Tools
Git, ImageJ, Power BI, Adobe Suite

Languages: English (Native) · French (Conversational)